Intro to Altair

SI 649 / EECS 548

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Adapted from Vega-Altair Documentation and Dallas Card

Plan for today

- Part 1:
  - Overview of Altair with Demo
- Part 2:
  - Recreate 1 visualization from NYT article, and create 1 more unique visualization
- Part 3:
  - Sharing progress with peers and discussion of assignment part 2

Altair

Three core pieces of Altair include:

- Data (Pandas DataFrame or other data source)
- Encodings (mapping properties of data to visual properties)
- Marks (line, bar, point, etc)

Data

Read more about types of data Altair accepts in documentation

Input data formats:

- Pandas DataFrame - Altair automatically detects data types of columns
- A URL pointing JSON/CSV files - User needs to manually identify column data types, e.g. ‘city:N’
- Data objects like alt.Data() for inline JSON-style data
- GeoPandas GeoDataFrame for geographic/spatial data

Encoding

Read about encoding and different channel options in [documentation]

Key encoding channels:

- Position: x, y
- Mark properties: color, shape, size, opacity
- Text: text, tooltip

Encoding data types:

- Q - quantitative (continuous real-valued quantity)
- O - ordinal (discrete ordered quantity)
- N - nominal (discrete unordered category)
- T - temporal (time or date)
- G - geojson (geographic shape)

Marks

Check out examples for each mark type in documentation

Marks specify how encoded attributes should appear on the plot. Altair supports the following primitive mark types:

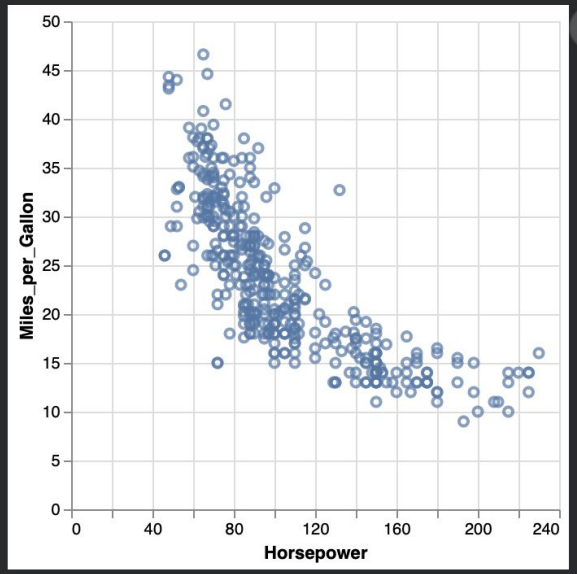
Mark	Method	Description
Arc	`mark_arc()`	A pie chart.
Area	`mark_area()`	A filled area plot.
Bar	`mark_bar()`	A bar plot.
Circle	`mark_circle()`	A scatter plot with filled circles.
Geoshape	`mark_geoshape()`	Visualization containing spatial data
Image	`mark_image()`	A scatter plot with image markers.
Line	`mark_line()`	A line plot.
Point	`mark_point()`	A scatter plot with configurable point shapes.
Rect	`mark_rect()`	A filled rectangle, used for heatmaps
Rule	`mark_rule()`	A vertical or horizontal line spanning the axis.
Square	`mark_square()`	A scatter plot with filled squares.
Text	`mark_text()`	A scatter plot with points represented by text.
Tick	`mark_tick()`	A vertical or horizontal tick mark.
Trail	`mark_trail()`	A line with variable widths.

```python
import altair as alt
from vega_datasets import data

cars = data.cars()

alt.Chart(cars).mark_point().encode(
    x='Horsepower',
    y='Miles_per_Gallon'
)
```

Data

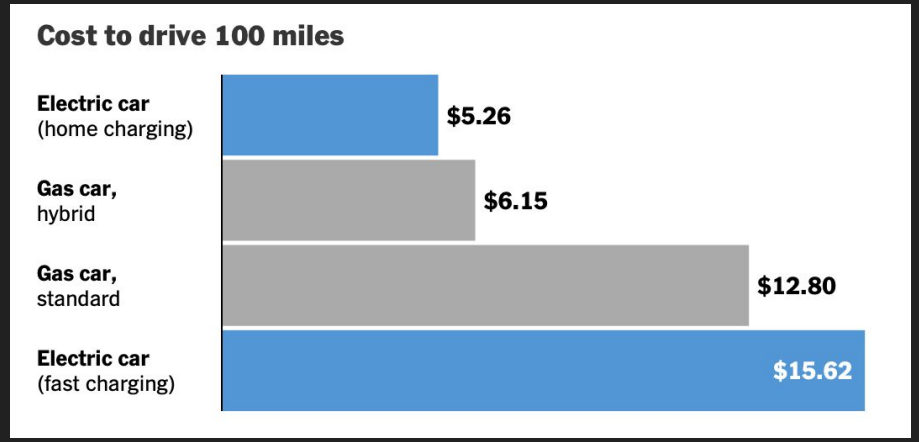
Mark

Encoding



Data provides the source (cars) → Mark defines the shape (points) → Encoding maps data columns to visual properties (x, y)

Altair demonstration

Lab 4 Assignment

- Make a copy of this notebook
- Take a look at the paper “How Much It Costs to Drive an E.V. and a Gas Car in Every State”
  - https://www.nytimes.com/2025/10/08/upshot/ev-versus-gas-car-comparison-cost.html
- Using Altair, recreate ‘Cost to drive 100 miles’ figure
- Try to get all stylistic details as close as possible
- Then, come up with a new visualization using the same data that can help us get more insight
- Use different mark type for your visualization